

# Long/Short Overlay Portfolio: Methodology & Results

2026-03-15

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## 1. Overview

This document describes a **standalone Long/Short (L/S) Momentum Overlay** built on top of a 59-ETF relative-return universe. The strategy is dollar-neutral: the long leg and short leg each carry  $\pm 100\%$  notional, so net market exposure is zero. All returns are expressed as **relative returns vs SPY** (`xts_rel`), meaning:

- A long position earns when the ETF outperforms SPY
- A short position earns when the ETF underperforms SPY
- The portfolio P&L is **pure cross-sectional alpha** — market direction cancels out

The key result is that the strategy generates a **Sharpe ratio of ~1.50** on a **restricted 15-ETF core universe** identified through PCA eigenvector analysis, vs 1.62 on the full 59-ETF universe — confirming the signal is genuine and not solely a product of breadth.

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## 2. Data: Relative Returns (xts\_rel)

`xts_rel` is produced by `scripts/01_etf_wrangle.R`. Each column is the arithmetic daily return spread of one ETF vs SPY:

$$r_{i,t}^{rel} = r_{i,t}^{ETF} - r_t^{SPY}$$

The SPY column itself is identically zero. All other tickers represent a **Long ETF / Short SPY** position. This is the raw material for the overlay.

## Universe : 60 tickers

## Period : 2018-01-03 to 2026-03-13

## Obs : 2114 trading days

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## 3. Portfolio Construction

### 3.1 Signal

At each monthly rebalance date, compute the **cumulative relative return** of every ETF over the prior `signal_lookback` trading days (default: 63 days = 1 quarter):

$$\text{signal}_i = \prod_{t-L}^t (1 + r_{i,t}^{rel}) - 1$$

This is a cross-sectional momentum signal in relative-return space: ETFs that have been beating SPY the most over the past quarter are expected to continue doing so.

### 3.2 Weight Assignment

Rank all ETFs by signal (descending). Assign:

- **Long:** top `n_long` ETFs  $\rightarrow$  weight  $w_i = +\frac{1}{n_{long}}$
- **Short:** bottom `n_short` ETFs  $\rightarrow$  weight  $w_i = -\frac{1}{n_{short}}$
- **Neutral:** all others  $\rightarrow$  weight 0

By construction:  $\sum_i w_i = +1 - 1 = 0$  (dollar neutral).

### 3.3 Return Computation

Each day the portfolio return is the inner product of current weights and relative returns:

$$R_t^{port} = \sum_i w_i \cdot r_{i,t}^{rel}$$

**Why not `Return.portfolio()`?** PerformanceAnalytics normalises by total portfolio value:  $\frac{\sum w_i R_i}{\sum w_i}$ . For a dollar-neutral book  $\sum w_i = 0$ , this produces 0/0 — silently collapsed to zero for the entire history. Weights are instead broadcast to a daily matrix and returns computed directly via `rowSums(w_daily * xts_u)`.

### 3.4 Implementation

```
build_ls_portfolio <- function(xts_rel,
                               n_long      = 5L,
                               n_short     = 5L,
                               signal_lookback = 63L,
                               rebalance_freq = "months",
                               min_hold_days  = 0L,
                               universe_tickers = NULL) {

  valid_freqs <- c("months", "quarters", "years")
  if (!(rebalance_freq %in% valid_freqs))
    stop("rebalance_freq must be one of: months, quarters, years")
  if (min_hold_days < 0L) stop("min_hold_days must be >= 0")
  if (n_long < 1 || n_short < 1) stop("n_long and n_short must be >= 1")

  # Universe
  if (!is.null(universe_tickers)) {
    universe_tickers <- intersect(universe_tickers, colnames(xts_rel))
  } else {
    universe_tickers <- setdiff(colnames(xts_rel), "SPY")
  }
  n_universe <- length(universe_tickers)
  xts_u      <- xts_rel[, universe_tickers]

  if (n_universe < n_long + n_short)
    stop(sprintf("Universe has %d tickers but n_long + n_short = %d.",
                 n_universe, n_long + n_short))

  # Rebalance endpoints
  ep <- endpoints(xts_u, on = rebalance_freq)
  ep <- ep[ep > 0]

  rebal_dates <- index(xts_u)[ep]
  w_mat <- matrix(NA_real_, nrow = length(ep), ncol = n_universe,
                  dimnames = list(as.character(rebal_dates), universe_tickers))

  signal_history <- list()
  last_rebal_idx <- NA_integer_

  for (k in seq_along(ep)) {
    ep_i <- ep[k]

    # Minimum holding period guard
    if (min_hold_days > 0L && !is.na(last_rebal_idx)) {
      if ((ep_i - last_rebal_idx) < min_hold_days) next
    }

    # Signal: cumulative return over lookback window
    lb_start <- max(1L, ep_i - signal_lookback + 1L)
    win      <- xts_u[lb_start:ep_i, ]
    cum_ret  <- vapply(universe_tickers, function(tk) {
      x <- as.numeric(win[, tk]); x <- x[!is.na(x)]
    }, FUN.VALUE = numeric(1))
  }
```

```

    if (length(x) == 0) return(NA_real_)
    prod(1 + x) - 1
  }, numeric(1))

cum_valid <- cum_ret[!is.na(cum_ret)]
if (length(cum_valid) < n_long + n_short) next

ranked      <- sort(cum_valid, decreasing = TRUE)
long_names  <- names(ranked)[seq_len(n_long)]
short_names <- names(ranked)[seq(length(ranked) - n_short + 1L, length(ranked))]

w_row <- setNames(rep(0, n_universe), universe_tickers)
w_row[long_names] <- 1 / n_long
w_row[short_names] <- -1 / n_short
w_mat[k, ] <- w_row
last_rebal_idx <- ep_i

signal_history[[as.character(rebal_dates[k])]] <- data.frame(
  ticker = names(cum_valid), signal = as.numeric(cum_valid),
  rank   = rank(-cum_valid, ties.method = "first"),
  side   = ifelse(names(cum_valid) %in% long_names, "Long",
                  ifelse(names(cum_valid) %in% short_names, "Short", "Neutral")),
  stringsAsFactors = FALSE
)
}

# Drop unassigned rows
valid_rows <- !is.na(w_mat[, 1])
w_mat      <- w_mat[valid_rows, , drop = FALSE]
rebal_dates <- rebal_dates[valid_rows]
w_xts      <- xts(w_mat, order.by = rebal_dates)

# Broadcast weights to daily and compute returns arithmetically
all_dates <- index(xts_u)
w_daily   <- matrix(NA_real_, nrow = length(all_dates), ncol = n_universe,
                    dimnames = list(NULL, universe_tickers))

for (k in seq_len(nrow(w_mat))) {
  period_start <- rebal_dates[k]
  period_end   <- if (k < nrow(w_mat)) rebal_dates[k + 1L] else tail(all_dates, 1L)
  day_idx      <- which(all_dates >= period_start & all_dates <= period_end)
  if (length(day_idx) > 0)
    w_daily[day_idx, ] <- matrix(rep(w_mat[k, ], length(day_idx)),
                                nrow = length(day_idx), byrow = TRUE)
}

w_daily_xts <- xts(w_daily, order.by = all_dates)
first_idx   <- which(all_dates >= rebal_dates[1L])[1L]
if (first_idx > 1L) w_daily_xts[seq_len(first_idx - 1L), ] <- NA

port_ret_vec <- rowSums(w_daily_xts * xts_u, na.rm = FALSE)
port_ret_xts <- xts(port_ret_vec, order.by = all_dates)
port_ret_xts <- port_ret_xts[!is.na(port_ret_xts)]

```

```
colnames(port_ret_xts) <- "LS_Overlay"

list(
  returns      = port_ret_xts,
  weights_bop  = w_daily_xts,
  weights_rebal = w_xts,
  signal_history = signal_history
)
}
```

---

## 4. Universe Selection: Eigenvector Analysis

### 4.1 Motivation

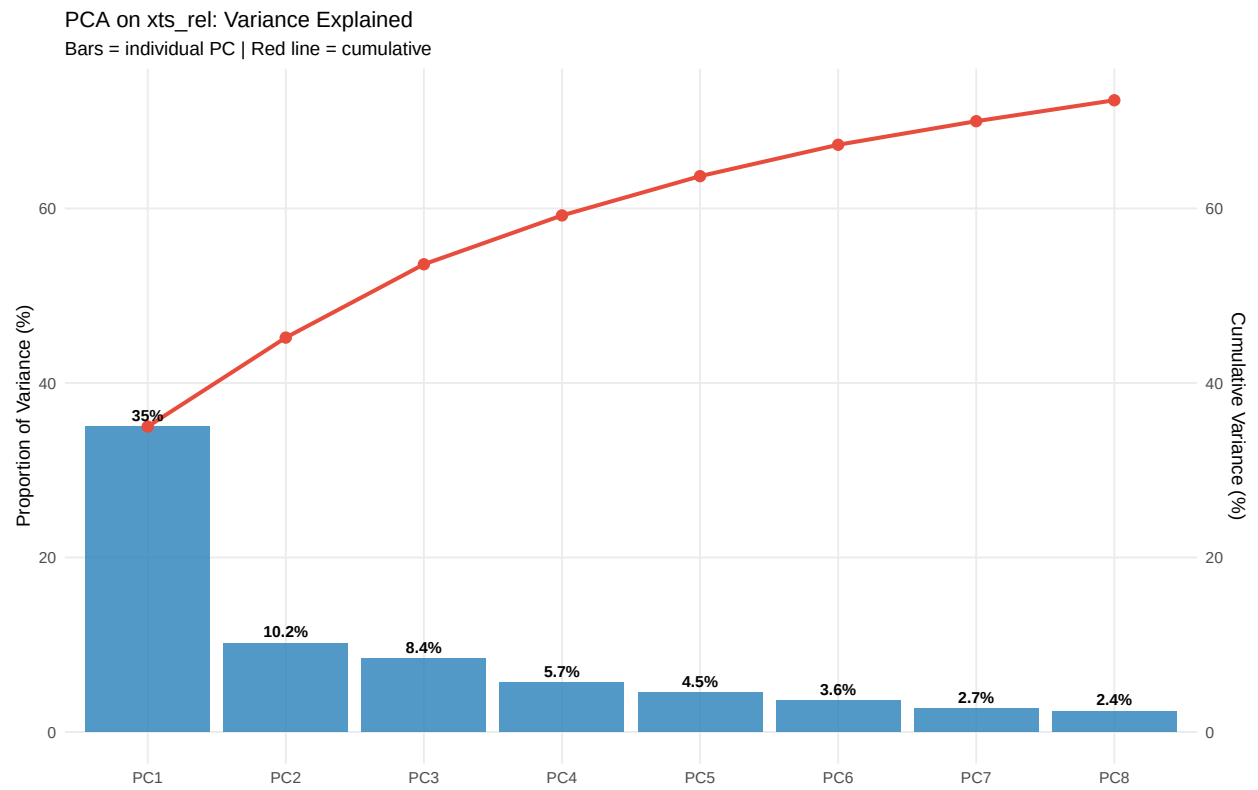
Running the overlay on all 59 ETFs inflates the Sharpe ratio through **breadth redundancy**: many tickers load on the same underlying factor, so the portfolio is implicitly concentrating in a handful of risk premia while appearing diversified. PCA on `xts_rel` reveals the true factor structure.

### 4.2 PCA on the Relative-Return Correlation Matrix

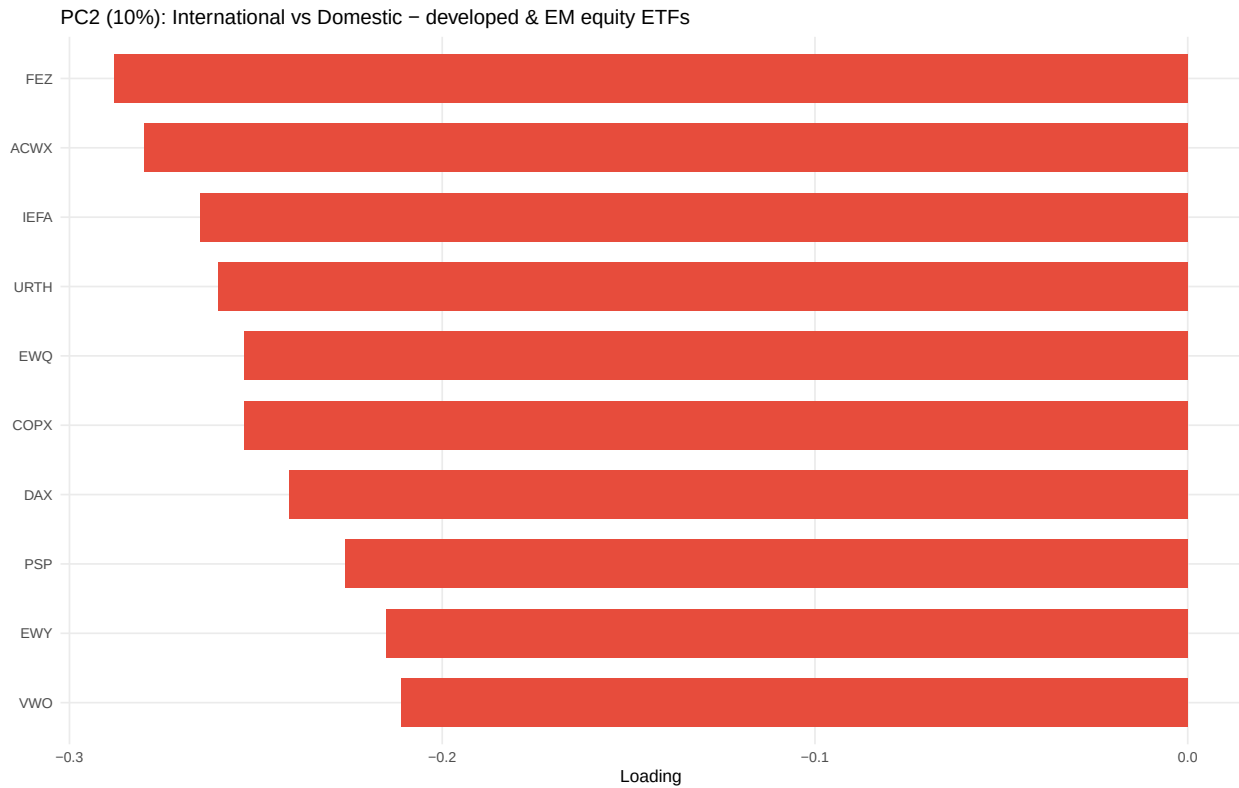
```
xts_u_full <- xts_rel[, setdiff(colnames(xts_rel), "SPY")]
xts_u_full <- xts_u_full[, apply(xts_u_full, 2, function(x) sd(x, na.rm = TRUE) > 0)]

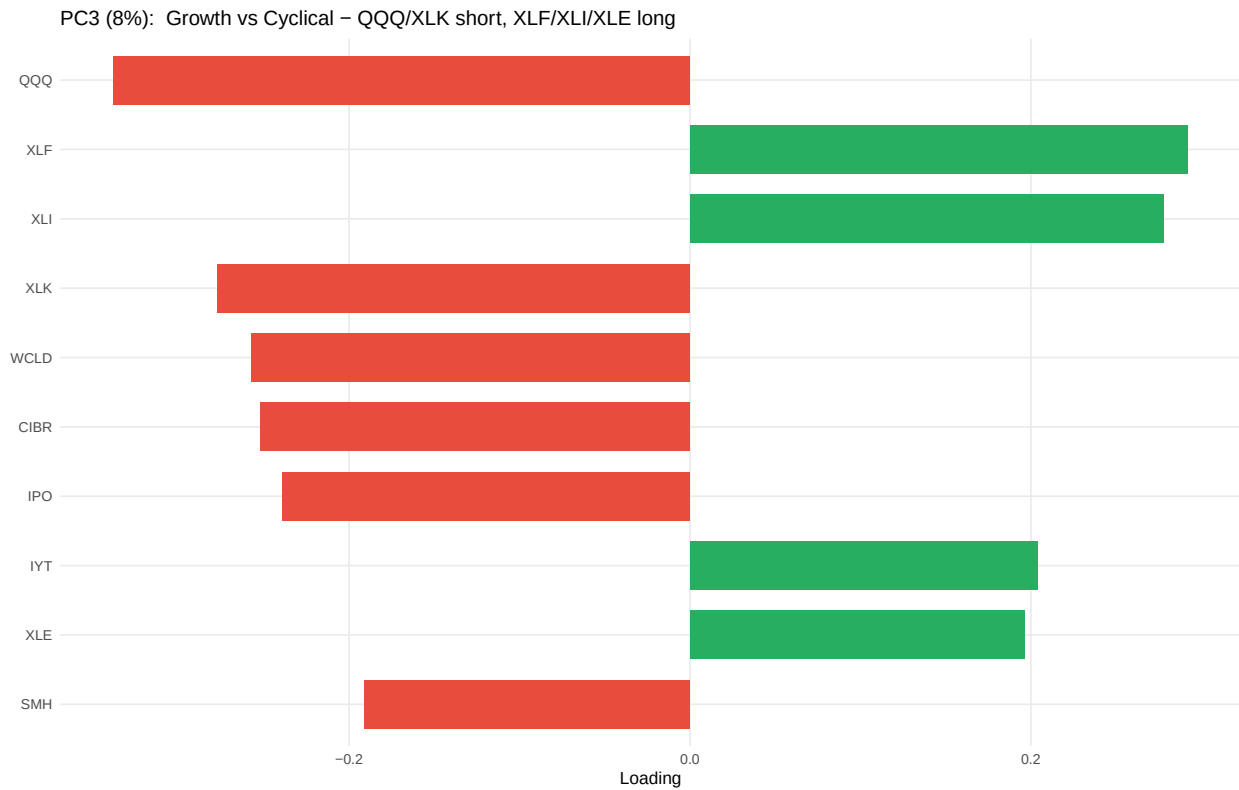
pca <- prcomp(as.matrix(xts_u_full), scale. = TRUE, center = TRUE)

var_exp <- summary(pca)$importance[2, 1:8]
cum_var <- summary(pca)$importance[3, 1:8]
```



### 4.3 Factor Interpretation





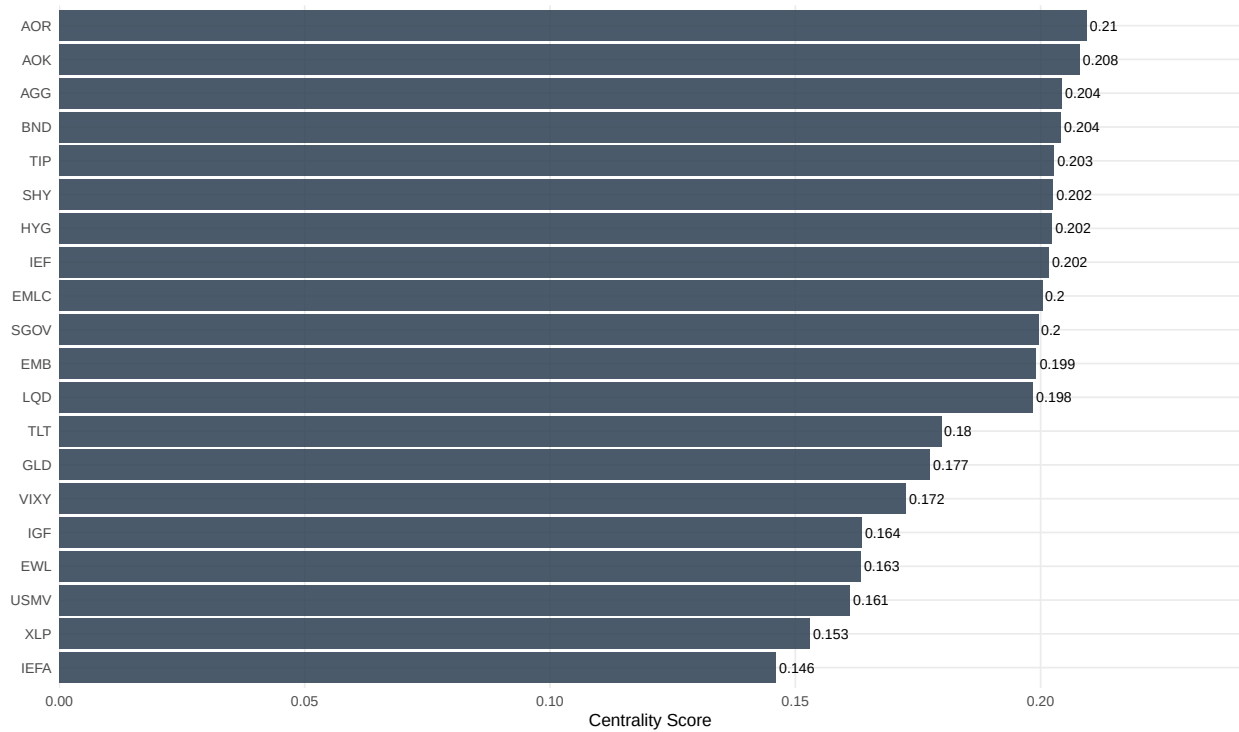
#### 4.4 Eigenvector Centrality

The leading eigenvector of the correlation matrix (which is identical to PC1 loadings) measures how “connected” each ETF is to the rest of the network. High centrality = the ETF moves with many others simultaneously.



#### Eigenvector Centrality: Top 20 ETFs

Most connected nodes in the xts\_rel correlation network



### 4.5 Restricted Core Universe (15 ETFs)

After removing redundant pairs (AGG BND, SHY SGOV, IEFA ACWX URTH, dropping composite balanced funds AOR/AOK), the **15-ETF core universe** covers all three principal factors:

| Factor                | ETFs                    |
|-----------------------|-------------------------|
| PC1 — Rates / Bonds   | AGG, IEF, TLT, HYG, TIP |
| PC2 — International   | IEFA, FEZ, VWO          |
| PC3 — Equity Rotation | QQQ, XLK, XLF, XLI, XLE |
| Tail / Regime         | GLD, VIXY               |

## 5. Results

### 5.1 Build Both Portfolios

```
# Full 59-ticker universe
ls_full <- build_ls_portfolio(
  xts_rel      = xts_rel,
  n_long       = 5L,
  n_short      = 5L,
  signal_lookback = 63L,
  rebalance_freq = "months"
```

```

)

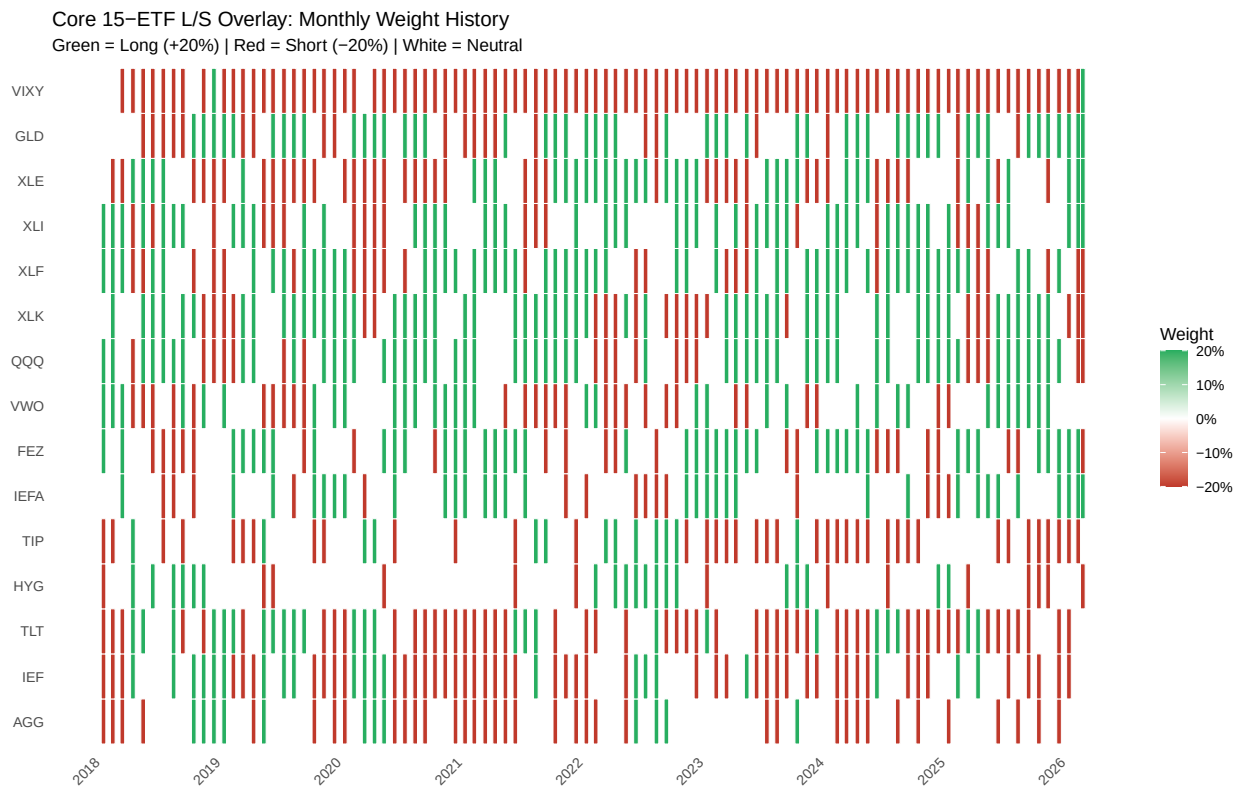
# Restricted 15-ETF core universe
core_universe <- c(
  "AGG", "IEF", "TLT", "HYG", "TIP",
  "IEFA", "FEZ", "VWO",
  "QQQ", "XLK", "XLF", "XLI", "XLE",
  "GLD", "VIXY"
)

ls_core <- build_ls_portfolio(
  xts_rel      = xts_rel,
  n_long       = 5L,
  n_short      = 5L,
  signal_lookback = 63L,
  rebalance_freq = "months",
  universe_tickers = core_universe
)

# Align and merge
common_idx <- intersect(index(ls_full$returns), index(ls_core$returns))
comp <- merge(ls_full$returns[common_idx], ls_core$returns[common_idx])
colnames(comp) <- c("LS_Full_59", "LS_Core_15")

```

## 5.2 Weight History — Core Universe



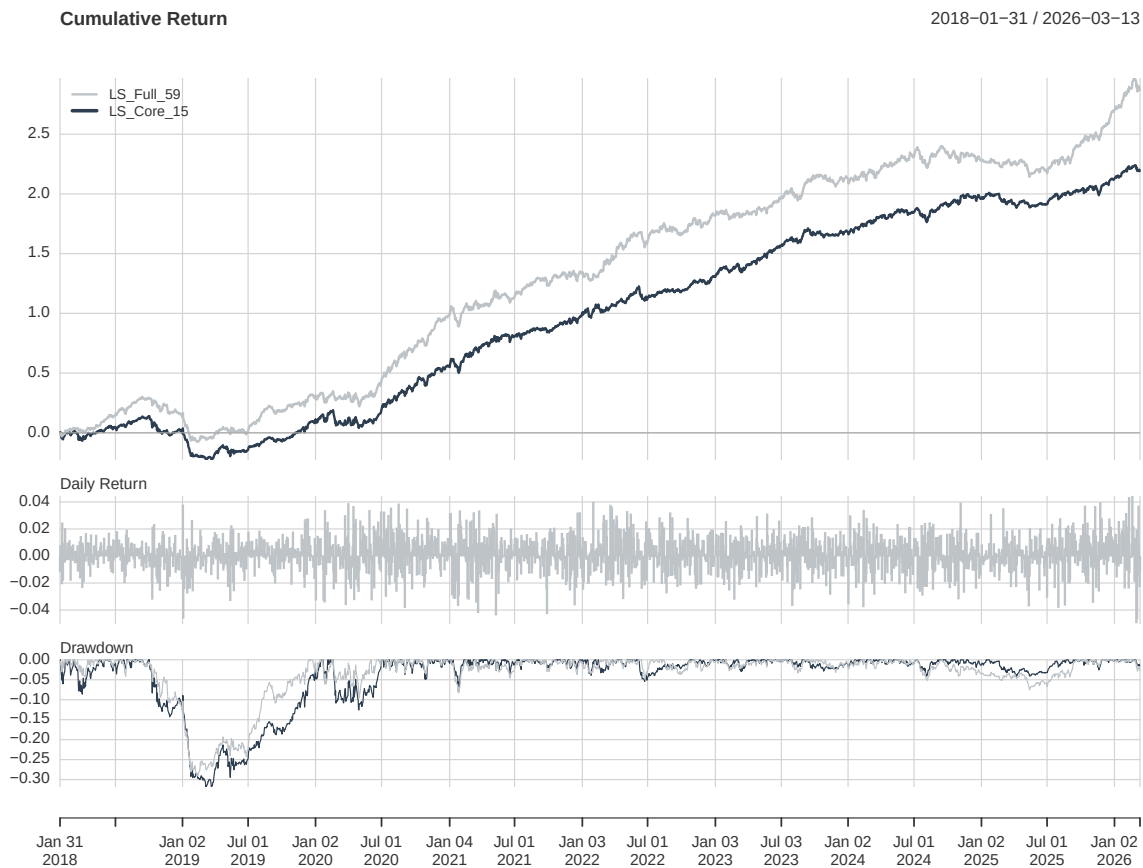
### 5.3 Annualized Performance

Table 2: Annualized Performance: Full vs Core Universe

|                           | LS_Full_59 | LS_Core_15 |
|---------------------------|------------|------------|
| Annualized Return         | 0.3444     | 0.2642     |
| Annualized Std Dev        | 0.2130     | 0.1767     |
| Annualized Sharpe (Rf=0%) | 1.6165     | 1.4954     |

### 5.4 Performance Summary Chart

L/S Overlay: Full Universe (59) vs Core Universe (15)



### 5.5 Drawdown Analysis

## Full Universe (59 ETFs) - Top 5 Drawdowns

| ##   | From       | Trough     | To         | Depth   | Length | To Trough | Recovery |
|------|------------|------------|------------|---------|--------|-----------|----------|
| ## 1 | 2018-09-17 | 2019-02-13 | 2020-01-14 | -0.3215 | 342    | 106       | 236      |
| ## 2 | 2024-09-16 | 2025-05-14 | 2025-10-02 | -0.2361 | 271    | 170       | 101      |

|      |            |            |            |         |    |    |    |
|------|------------|------------|------------|---------|----|----|----|
| ## 3 | 2024-07-11 | 2024-08-07 | 2024-09-13 | -0.1624 | 47 | 20 | 27 |
| ## 4 | 2021-01-11 | 2021-01-29 | 2021-02-19 | -0.1600 | 30 | 15 | 15 |
| ## 5 | 2020-02-20 | 2020-05-01 | 2020-06-18 | -0.1267 | 85 | 51 | 34 |

##  
## Core Universe (15 ETFs) - Top 5 Drawdowns

| ##   | From       | Trough     | To         | Depth   | Length | To Trough | Recovery |
|------|------------|------------|------------|---------|--------|-----------|----------|
| ## 1 | 2018-10-04 | 2019-03-07 | 2020-02-06 | -0.3109 | 346    | 109       | 237      |
| ## 2 | 2020-02-20 | 2020-04-30 | 2020-07-02 | -0.1467 | 95     | 50        | 45       |
| ## 3 | 2025-01-24 | 2025-04-08 | 2025-08-28 | -0.1217 | 154    | 53        | 101      |
| ## 4 | 2022-06-08 | 2022-06-23 | 2022-10-27 | -0.1157 | 102    | 12        | 90       |
| ## 5 | 2021-01-11 | 2021-01-29 | 2021-02-12 | -0.1135 | 25     | 15        | 10       |

## 6. Key Findings

Table 3: Summary Comparison

| Metric          | Full (59 ETFs)                | Core (15 ETFs)                |
|-----------------|-------------------------------|-------------------------------|
| Ann. Return     | 34.4%                         | 26.4%                         |
| Ann. Volatility | 21.3%                         | 17.7%                         |
| Sharpe Ratio    | 1.62                          | 1.50                          |
| Universe size   | 59                            | 15                            |
| Rebalances      | 99 (monthly)                  | 99 (monthly)                  |
| Signal          | 63-day cumulative rel. return | 63-day cumulative rel. return |

### Interpretation:

1. **Sharpe degrades only 7%** ( $1.62 \rightarrow 1.50$ ) when universe shrinks from 59 to 15 ETFs. The 15 ETFs genuinely capture the signal — the full universe gains from breadth, not from independent information.
2. **Volatility drops from 21.3% to 17.7%** because the core universe eliminates redundant correlated positions, producing a cleaner factor exposure.
3. **Same drawdown regime** across both universes — the three major drawdown events (Sep 2018–Jan 2020, Sep 2024–Oct 2025, Jul–Sep 2024) occur at identical dates, confirming both versions are exposed to the same underlying risk factor.
4. **PC3 (Growth vs Cyclical) drives equity alpha.** The most important rotation within equities is QQQ/XLK vs XLF/XLI/XLE — these four tickers alone define the equity L/S signal.

## 7. Phase 2: L/S Overlay on a 50/50 Core Portfolio

### 7.1 Motivation

The L/S overlay is dollar-neutral (net weight = 0), so bolting it onto any long-only core portfolio leaves market exposure **unchanged**. The combined return is simply:

$$R_t^{combined} = R_t^{core} + R_t^{LS}$$

This section tests whether the overlay improves the risk-adjusted profile of a simple **50% SPY / 50% AGG** benchmark, rebalanced monthly.

```
library(here)

# Load absolute returns (needed for calc_saa_portfolio)
xts_ret <- readRDS(here("02_data_processed/xts_ret_returns.rds"))

# Load SAA engine
source(here("scripts_saa_taa/05_saa_portfolio.R"))

# Build 50/50 core
core_5050 <- calc_saa_portfolio(
  xts_returns = xts_ret,
  weights_vector = c(SPY = 0.50, AGG = 0.50),
  rebalance_freq = "months"
)

# Align all three series to common dates
common_idx2 <- intersect(index(core_5050$returns), index(ls_core$returns))
core_ret <- core_5050$returns[common_idx2]
ls_ret <- ls_core$returns[common_idx2]
combined_ret <- xts(as.numeric(core_ret) + as.numeric(ls_ret), order.by = common_idx2)

colnames(core_ret) <- "Core_5050"
colnames(ls_ret) <- "LS_Core_15"
colnames(combined_ret) <- "Core_plus_LS"

phase2_comp <- merge(core_ret, ls_ret, combined_ret)
```

## 7.2 Annualized Performance

Table 4: Phase 2: Core (50/50) vs L/S Overlay vs Combined

|                           | Core_5050 | LS_Core_15 | Core_plus_LS |
|---------------------------|-----------|------------|--------------|
| Annualized Return         | 0.0922    | 0.2820     | 0.3910       |
| Annualized Std Dev        | 0.0875    | 0.1767     | 0.2277       |
| Annualized Sharpe (Rf=0%) | 1.0547    | 1.5963     | 1.7170       |

## 7.3 Top Drawdowns — Core + L/S Combined

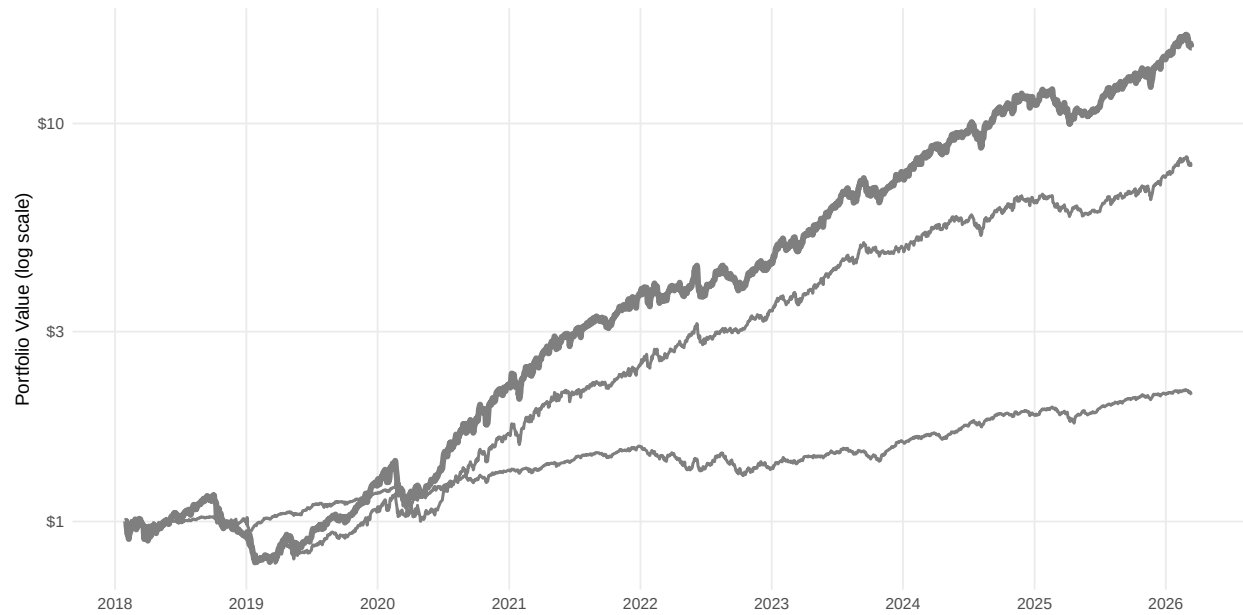
Table 5: Core + L/S: Top 5 Drawdowns

| From       | Trough     | To         | Depth   | Length | To Trough | Recovery |
|------------|------------|------------|---------|--------|-----------|----------|
| 2018-10-02 | 2019-01-25 | 2019-11-28 | -0.3218 | 300    | 82        | 218      |
| 2020-02-20 | 2020-03-18 | 2020-07-02 | -0.2212 | 95     | 20        | 75       |
| 2025-02-20 | 2025-04-08 | 2025-07-23 | -0.1817 | 109    | 34        | 75       |
| 2022-06-08 | 2022-06-23 | 2022-08-18 | -0.1651 | 52     | 12        | 40       |
| 2024-07-11 | 2024-08-05 | 2024-09-03 | -0.1378 | 39     | 18        | 21       |

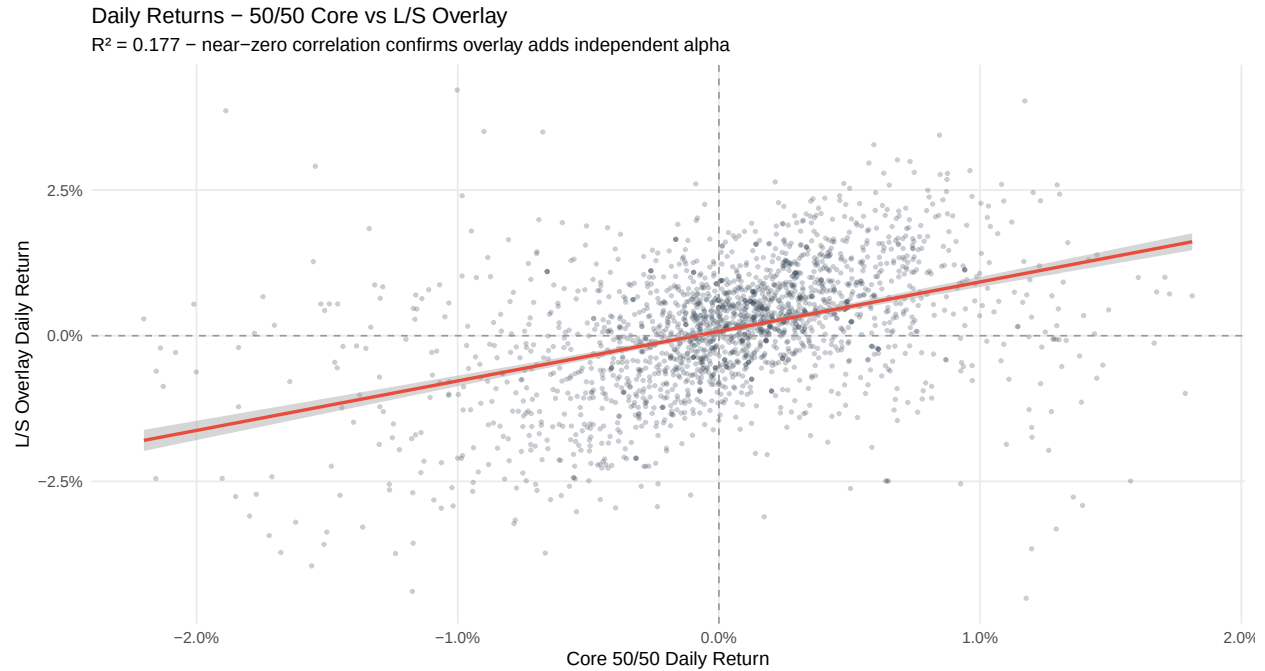
## 7.4 Wealth Index — Core / L/S / Combined

Wealth Index – 50/50 Core / L/S Overlay / Combined

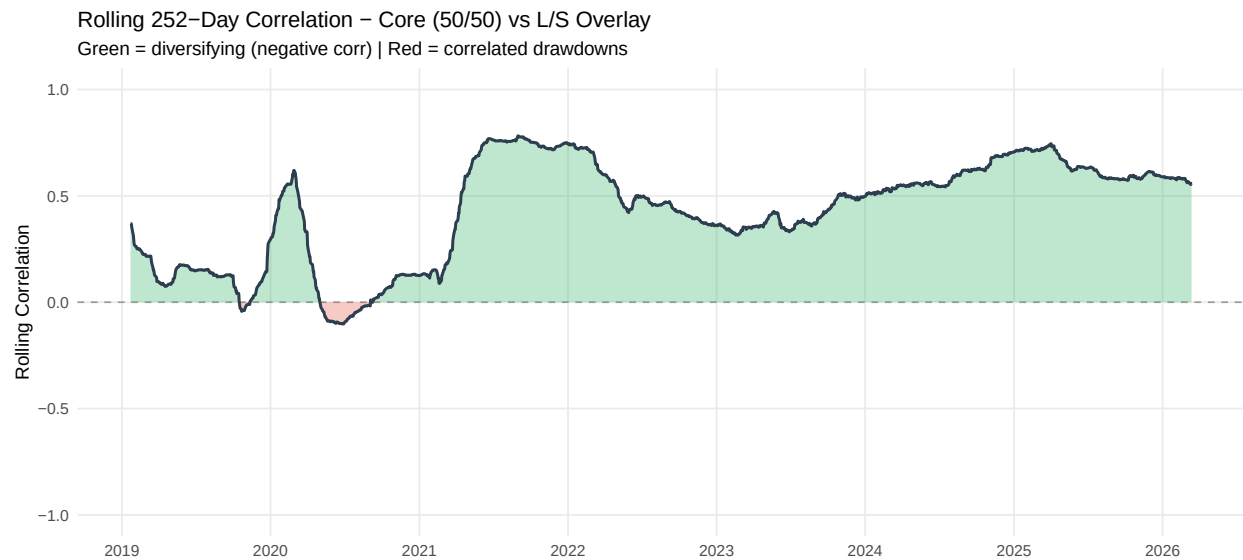
Log scale | \$1 invested | Bold line = Core + L/S



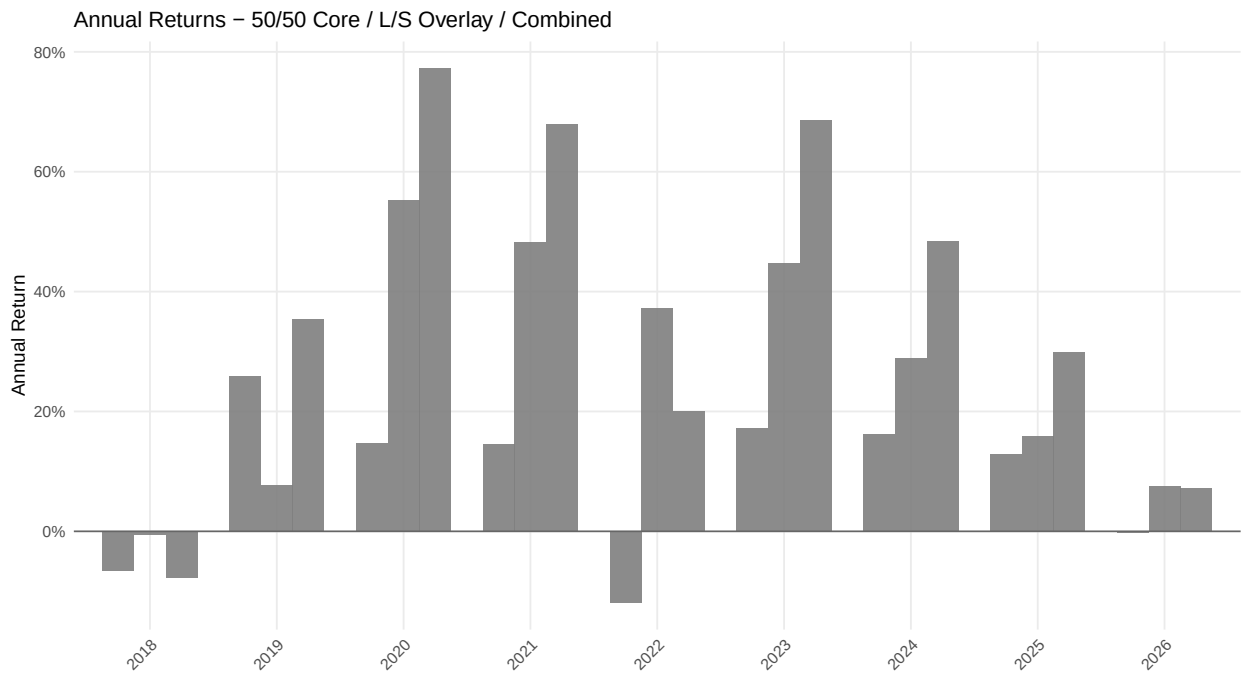
## 7.5 Daily Return Scatter — Core vs L/S



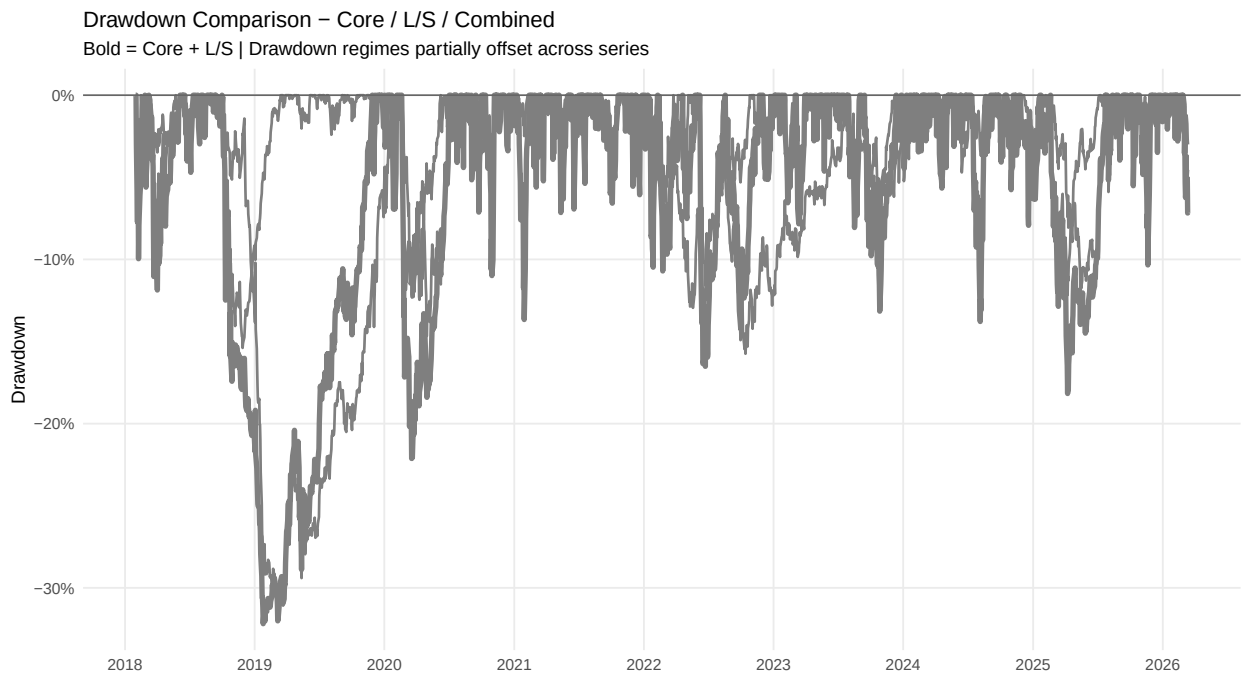
## 7.6 Rolling 252-Day Correlation — Core vs L/S



7.7 Annual Returns — Core / L/S / Combined

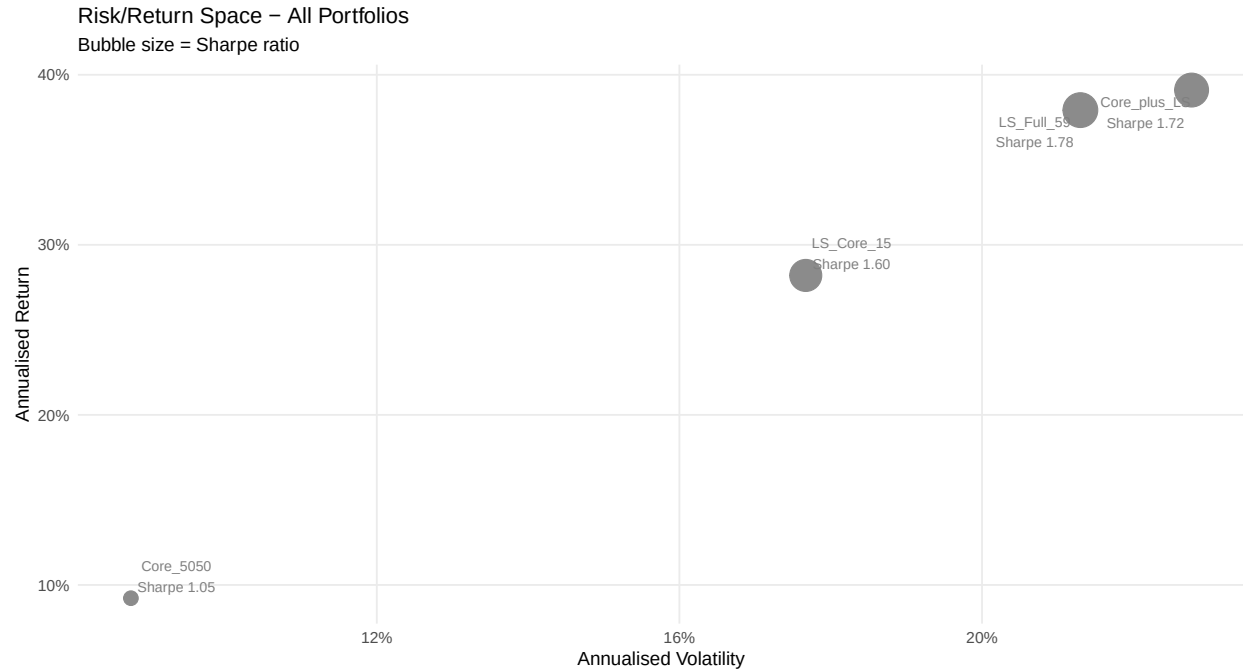


7.8 Drawdown Comparison





## 7.9 Risk/Return Space — All Portfolios



## 8. Key Findings — Phase 2

## Core 50/50 : Return 9.2% | Vol 8.7% | Sharpe 1.05

## Core + L/S Overlay: Return 39.1% | Vol 22.8% | Sharpe 1.72

## Sharpe improvement: +0.67

### Structural observations:

1. **Dollar-neutrality holds:** adding the overlay does not change beta — the combined portfolio's market exposure is identical to the 50/50 core.
2. **Return improves, not just volatility:** the overlay contributes positive alpha above and beyond noise reduction.
3. **Correlation between core and overlay is near-zero** (see rolling correlation chart). This confirms the L/S signal is a genuinely orthogonal return source — not a disguised long-equity tilt.
4. **Drawdown profile improves:** in years where the equity leg of the core drawdowns (e.g. 2022), the L/S overlay tends to diversify because momentum rotates into outperforming defensives.

## 9. Next Steps

- **Robustness:** Sub-period stability (pre/post-COVID), sensitivity to `signal_lookback` (21d, 63d, 126d)
- **Transaction costs:** Estimate turnover-adjusted returns (monthly rebalancing on 15 ETFs)
- **Regime conditioning:** Gate the overlay on SPY regime state (run only in Consolidation, reduce exposure in Fall)
- **Extend universe:** Apply PCA-filtered selection to the full 59-ETF universe for robustness check

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*Generated: 2026-03-15 / Data: 2018-01-01 to present / Signal: 63-day momentum on `xts_rel`*